

Power Measurement in 3- ϕ circuits:- (Important) (2)

UNIT-3

Introduction:-

- In AC circuits, power is measured with the help of a wattmeter.
- A wattmeter is an Instrument that consist of two coils called current coil & potential coil.
- The current coil having low Resistance is connected in series with the load, so that it carries the load current.
- The potential coil have high Resistance is connected across the load & carries the current proportional to the potential difference across the load.
- for measuring power in a polyphase system, more than one wattmeter is Required or more than one Reading is made by one wattmeter.
- The 1st method is more convenient & the number of wattmeter Required to Measure power in a given poly-phase system is determined by Blondel's theorem.
- According to Blondel, a french engineer, when power is supplied by k -wire AC system, the number of wattmeter Required to Measure power is one less than the number of wires, that is $(k-1)$, ^{regardless the} load is Balanced or unbalanced.

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Three-Wattmeter Method:-

→ Hence, three wattmeters are Required to Measure Power in 3- ϕ , 4 wire system, while only two wattmeters are Required to measure Power in 3- ϕ , 3 wire system.

Three - Wattmeter Method:-

→ This Method is employed to Measure power in 3- ϕ , 4-wire system. However, this Method can also be employed in 3- ϕ , 3-wire delta connected load, where power consumed by each load is Required to be determined Separately.

→ The connections for both Star & delta Connected load are shown in figure below:-

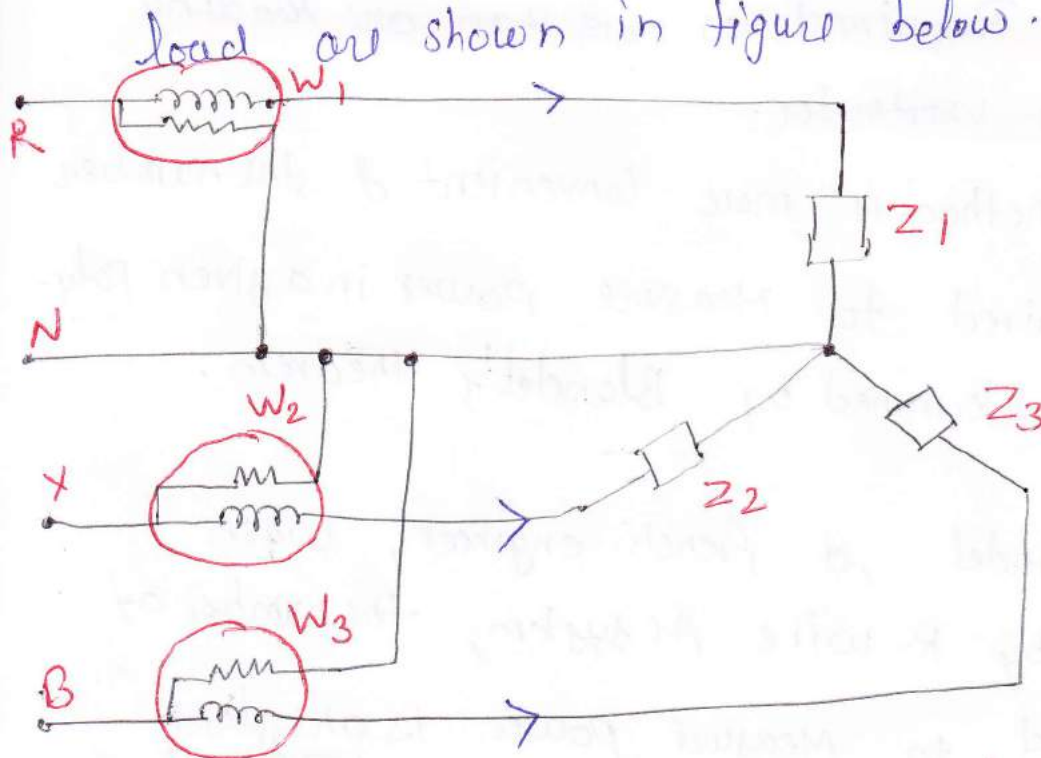


fig → connections of 3-wattmeter for power measurement in star connected loads.

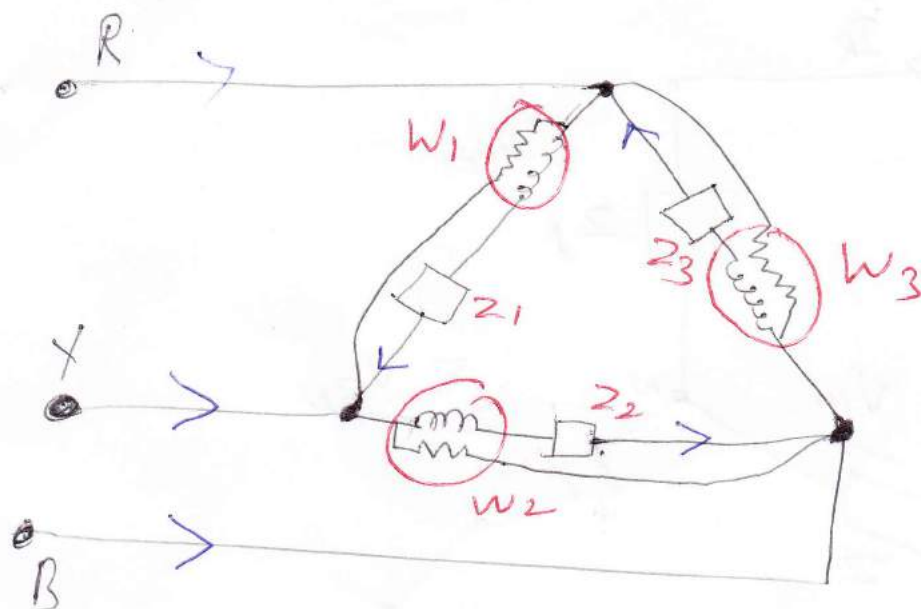


fig → Connection of three-wattmeters for power Measurements in delta connected load.

→ The total power is given by the algebraic sum of Reading of three wattmeter.

$$\text{Total power, } P = W_1 + W_2 + W_3$$

Except for three-phase, ~~four~~ four-wire unbalanced loads, three- ϕ power can be measured by using only two wattmeters.

⊗ Two-Wattmeter Method:- (Balanced load) (V. Important)

→ In this case, we shall prove that Power measured by the two wattmeters (i.e. sum of two wattmeter Reading) is equal to $\sqrt{3} V_L I_L \cos \phi$, which is the actual power consumed in a 3- ϕ Balanced load.

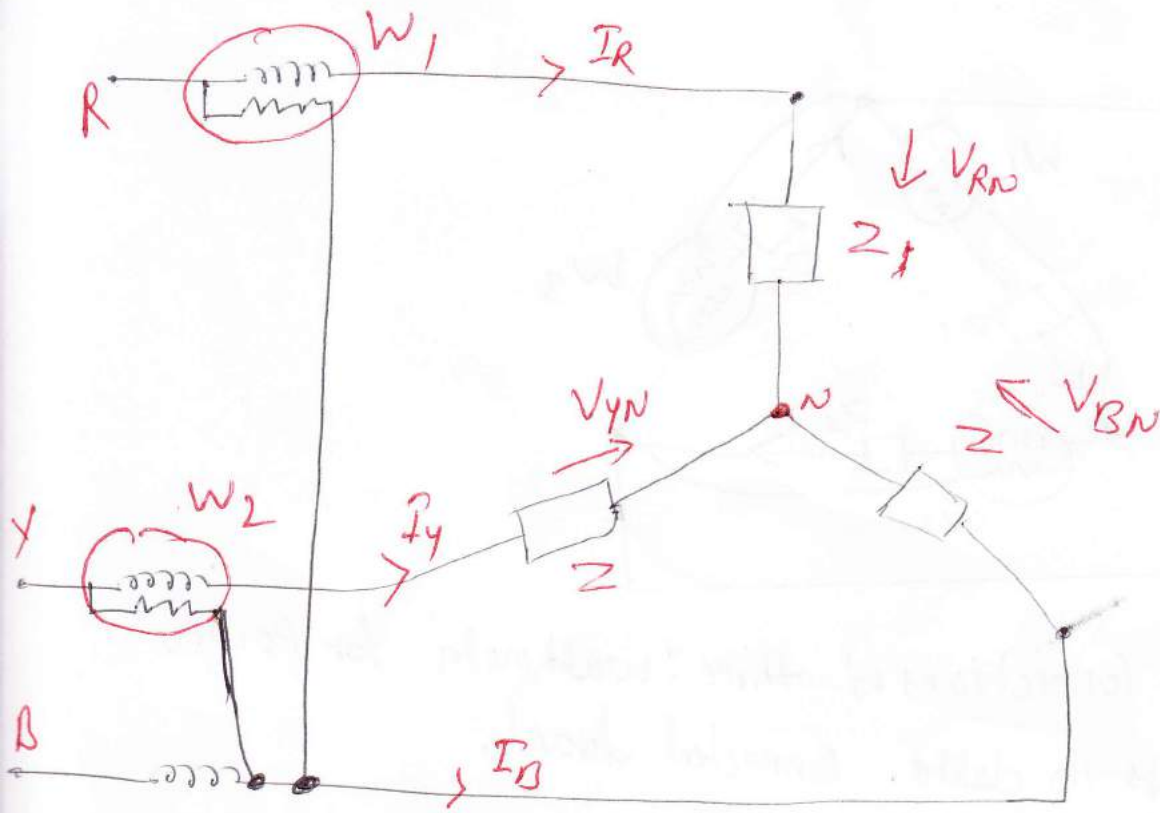


fig → connection of two wattmeters for power measurement in balanced 3- ϕ load.

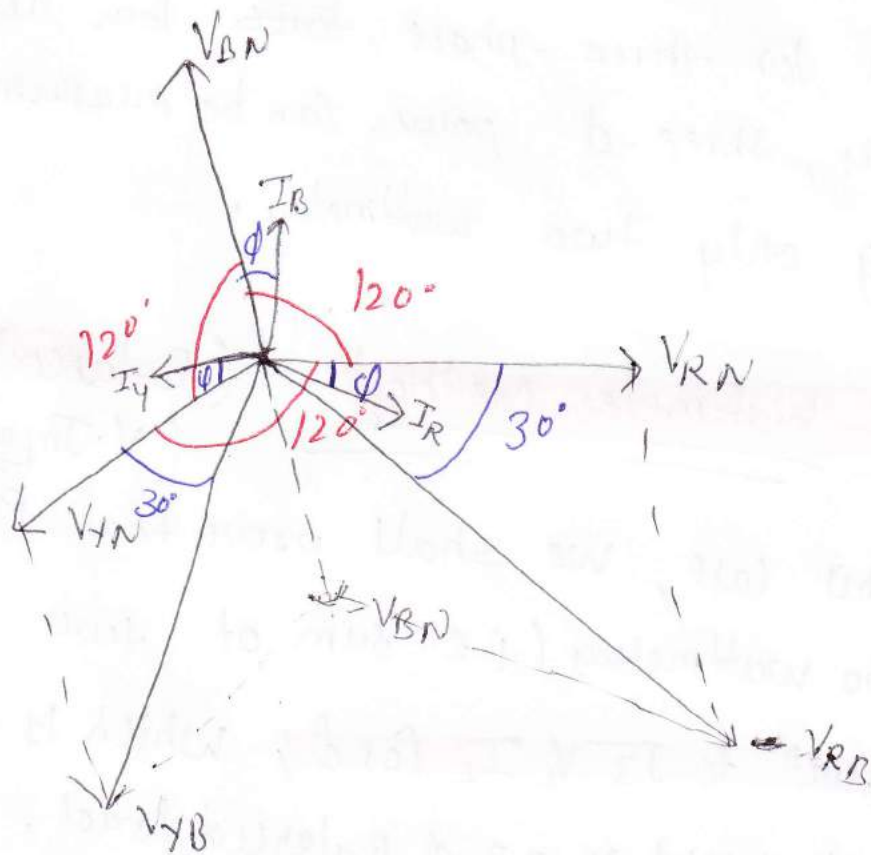


fig → phasor diagram

→ The Connection diagram for a 3- ϕ Balanced load ⁽²⁵⁾ is shown in fig. (Page 24)

→ Considering load to be an inductive one, the phasor diagram is shown in fig (Page 24)

→ The three-phase voltages V_{RN} , V_{YN} & V_{BN} displaced by an angle of 120° electrical are shown in phasor diagram.

→ The phase currents lag behind their respective phase voltages by an angle ϕ .

Current through current coil (C.C) of $W_1 = I_R$

Potential difference across potential coil (P.C) of W_1 is $\bar{V}_{RB} + \bar{V}_{BN} - \bar{V}_{RN} = 0 \rightarrow \boxed{W_1 = \bar{V}_{RB} = \bar{V}_{RN} = \bar{V}_{BN}}$ (using $\bar{V}_{BN}$)

To obtain V_{RB} , Reverse the phasor V_{BN} & add it vectorially to phasor V_{RN} , as shown in the phasor diagram (Page No. 24)

The phase difference b/w V_{RB} & I_R is $(30^\circ - \phi)$

Power Measured by wattmeter

$$\boxed{W_1 = V_{RB} I_R \cos(30^\circ - \phi)}$$

Current through l.c of $W_2 = I_Y$

Potential difference b/w V_{YB} & I_Y is $(30^\circ + \phi)$

Power Measured by Wattmeter, $W_2 = V_{YB} = V_{BR} = V_L$

Potential difference across p.c of $W_2 = \overline{V_{YB}} = \overline{V_{YN}} - \overline{V_{BN}}$

→ To obtain ~~the~~ V_{YB} , Reverse the phasor V_{BN} & add it vectorially to phasor V_{YN} as shown in phasor diagram (Page No. 24)

Wattmeter Reading, $W_1 = V_L I_L \cos(30^\circ - \phi)$

" " " $W_2 = V_L I_L \cos(30^\circ + \phi)$

Sum of the two wattmeter Reading

$$W_1 + W_2 = V_L I_L \cos(30^\circ - \phi) + V_L I_L \cos(30^\circ + \phi)$$

$$= V_L I_L [\cos(30^\circ - \phi) + \cos(30^\circ + \phi)]$$

$$\Rightarrow V_L I_L [\cos 30^\circ \cos \phi + \sin 30^\circ \sin \phi + \cos 30^\circ \cos \phi - \sin 30^\circ \sin \phi]$$

$$= V_L I_L (2 \cos 30^\circ \cos \phi) = V_L I_L \left[2 \times \frac{\sqrt{3}}{2} \times \cos \phi \right]$$

$$\boxed{P = \sqrt{3} V_L I_L \cos \phi} \quad (\text{total power absorbed by a 3-ph Balanced load (P)})$$

$$\boxed{P = W_1 + W_2}$$

(*) Determination of Power factor from Wattmeter (27)
Readings:-

We have seen that

$$\boxed{W_1 + W_2 = \sqrt{3} V_L I_L \cos \phi} \quad \text{--- eqn (1)}$$

$$\begin{aligned} W_1 - W_2 &= V_L I_L (\cos(30^\circ - \phi) - \cos(30^\circ + \phi)) \\ &= 2 V_L I_L \sin 30^\circ \sin \phi \end{aligned}$$

$$\boxed{W_1 - W_2 = V_L I_L \sin \phi} \quad \rightarrow \text{eqn (2)}$$

Dividing eqn (2) by (1)

$$\frac{W_1 - W_2}{W_1 + W_2} = \frac{V_L I_L \sin \phi}{\sqrt{3} V_L I_L \cos \phi}$$

$$\tan \phi = \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2} \Rightarrow \phi = \tan^{-1} \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2}$$

$$\cos \phi = \cos \tan^{-1} \phi$$

$$\boxed{\cos \phi = \cos \tan^{-1} \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2}}$$

Determination of Reactive Power from two wattmeter
Reading:-

Multiplying eqn (2) by $\sqrt{3}$, we get

$$\sqrt{3} (W_1 - W_2) = \sqrt{3} V_L I_L \sin \phi = P_r$$

$$P_r = \sqrt{3} (W_1 - W_2)$$

⊗ Effect of power factor on the two wattmeter (28)

Reading:-

(Important)

→ for lagging power factor, the wattmeter readings are

$$W_1 = V_L I_L \cos(30^\circ - \phi) \quad \& \quad W_2 = V_L I_L \cos(30^\circ + \phi)$$

→ It is clear that the Reading of the two wattmeter do not only depend upon the load, but they also depend upon the phase angle ϕ or the power factor of the load.

1) power factor is unity ($\cos \phi = 1$) or $\phi = 0^\circ$

for pure Resistance load, the power factor is unity, that is $\cos \phi = 1$ or $\phi = 0^\circ$

then the, two wattmeter Reading will be

$$W_1 = V_L I_L \cos(30^\circ - 0) = V_L I_L \cos 30^\circ$$

$$W_2 = V_L I_L \cos(30^\circ + 0) = V_L I_L \cos 30^\circ$$

$$W_1 = W_2$$

∴ Hence, When power factor is unity, both the wattmeters give equal Reading.

② power factor is 0.5 ($\cos\phi = 0.5$) or $\phi = 60^\circ$:- (29)

Under such condition, the two wattmeter Reading will be

$$W_1 = V_L I_L \cos(30^\circ - 60^\circ) = V_L I_L \cos 30^\circ$$

$$W_2 = V_L I_L \cos(30^\circ + 60^\circ) = 0$$

Hence, when the power factor is 0.5, one of the wattmeter W_2 gives zero reading (no deflection) and whole of the power measured by other wattmeter W_1 .

③ power factor is More than 0.5 but less than one
(i.e. $1 > \cos\phi > 0.5$) or ~~$60^\circ > \phi > 0^\circ$~~ $60^\circ > \phi > 0^\circ$

$$W_1 = V_L I_L \cos(30^\circ - \phi) \quad (+ve, \text{large})$$

$$W_2 = V_L I_L \cos(30^\circ + \phi) \quad (+ve, \text{smaller})$$

→ Hence, when the power factor is more than 0.5 but less than one, both the wattmeter gives +ve (upscale) Readings.

$$P = W_1 + W_2$$

= larger Reading + smaller Readings.

(4) Power factor is less than 0.5 but More than 0 (i.e. $0.5 > \cos \phi > 0$) or $60^\circ < \phi < 90^\circ$ (30)

$$W_1 = V_L I_L \cos(30^\circ - \phi) \quad (+ve, \text{larger})$$

$$W_2 = V_L I_L \cos(30^\circ + \phi) \quad (-ve, \text{smaller})$$

$W_2 \rightarrow$ gives -ve down scale Reading & W_1 gives +ve up scale Reading
 $P = W_1 + (-W_2)$

(5) Power factor is 0 ($\cos \phi = 0$) or $\phi = 90^\circ$

$$W_1 = V_L I_L \cos 60^\circ \quad (+ve)$$

$$W_2 = V_L I_L \cos 120^\circ$$

$$W_2 = -V_L I_L \cos 60^\circ \quad (-ve)$$

$$P = W_1 + W_2$$

$$P = W_1 + (-W_2)$$

$$P = W_1 - W_2 = 0$$

(It means, load is pure Inductive or Capacitive)

Note:-

(Do all the Numericals Related to the two-wattmeter Method from J.B. Gupta)

Numericals are also Important)

Numericals:-

(31)

Q:-1 In the two-wattmeter method of power measurement in a 3- ϕ circuit, the reading of the two wattmeters are 1000 W & 550 W. What is the power factor of the load?

Solⁿ

Reading of Wattmeter, $W_1 = 1000 \text{ W}$

" " " " , $W_2 = 550 \text{ W}$

$$\text{Power factor of load, } \cos \phi = \cos \tan^{-1} \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2}$$

$$\cos \phi = \cos \tan^{-1} \sqrt{3} \frac{1000 - 550}{1000 + 550} = 0.893 \text{ (lag)}$$

Q:-2 For a certain load, one of the wattmeter reads 20 kW & the other 5 kW after the voltage of this wattmeter has been reversed. Calculate the power & power factor of the load.

Solⁿ

$$W_1 = 20 \text{ kW}$$

$$W_2 = -5 \text{ kW}$$

$$P = W_1 + W_2 = 20 - 5 = 15 \text{ kW}$$

$$\cos \phi = \cos \tan^{-1} \sqrt{3} \frac{20 - (-5)}{20 + (-5)} = 0.3273 \text{ (lag)}$$

Q:-3 In a two-wattmeter method, power measured was 30 kW at 0.7 power factor lagging. Find the Reading of each wattmeter. (32)

Solⁿ

$$P = 30 \text{ kW}, \cos \phi = 0.7$$

$$P = W_1 + W_2 = 30 \text{ kW} \quad \text{--- eqⁿ ①}$$

$$\cos \phi = 0.7$$

$$\tan \phi = \tan (\cos^{-1}(0.7)) = 1.0202$$

$$\tan \phi = \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2}$$

$$1.0202 = \sqrt{3} \frac{W_1 - W_2}{30}$$

$$W_1 - W_2 = 17.67 \text{ kW} \quad \text{--- eqⁿ ②}$$

Solving eqⁿ (1) & (2)

$$W_1 = 23.835 \text{ kW} \text{ \& } W_2 = 6.165 \text{ kW}$$